

Performance of Alzheimer Disease Plasma Biomarkers in Patients With Prion Diseases

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Abstract

Background and Objectives

Prion diseases can mimic Alzheimer disease (AD) at presentation. Alzheimer's Association AD diagnostic criteria suggest that a single abnormal highly specific plasma biomarker (including p-tau217) is sufficient for a biological diagnosis. We investigated the performance of AD plasma biomarkers in distinguishing AD and prion diseases.

Methods

We examined plasma biomarker data from patients with prion disease from a prospective cohort study recruited through the UK National Prion Clinic. Prion diseases were diagnosed clinically or with autopsy confirmation, and AD was diagnosed clinically with CSF biomarker confirmation. Plasma p-tau217, p-tau181, Aβ42/40 ratio, brain-derived tau (BD-tau), neurofilament light chain (NfL), and glial fibrillary acid protein (GFAP) were measured using Simoa. Median biomarker values in different groups were compared with Kruskal-Wallis test, and area under the receiver operating characteristic curve was used to compare accuracy in distinguishing prion diseases from sporadic AD (sAD). Lumipulse p-tau217 and NfL were measured in a validation study in a different laboratory.

Results

In the main study, we analyzed 345 samples from 278 individuals (mean age 58 [SD 13.5], 48.2% female), including 204 with prion diseases (121 sporadic Creutzfeldt-Jakob disease [CJD], 11 iatrogenic CJD, 9 variant CJD, 47 slow-progressing inherited prion disease [IPD] and 16 fast-progressing IPD), 33 with AD, and 41 healthy controls. For discriminating prion disease without AD copathology from sAD, none of p-tau217 (area under the curve [AUC] [95% CI] 0.605 [0.486–0.724]), p-tau181 (AUC 0.554 [0.446–0.661]), or GFAP (AUC 0.514 [0.389–0.640]) performed well. Aβ42/40 discriminated moderately (AUC 0.770 [0.684–0.856]). NfL/p-tau217 ratio (AUC 0.996 [0.987–1.000]), NfL (AUC 0.988 [0.974–1.000]), BD-tau/p-tau217 ratio (AUC 0.963 [0.929–0.996]), and BD-tau (AUC 0.934 [0.890–0.978]) discriminated very well. In an independent validation study, consecutive samples were analyzed from 32 patients with sAD and 35 patients with sporadic Creutzfeldt-Jakob disease (mean age 65.0 [SD 6.4], 56.7% female). NfL/p-tau217 again discriminated almost perfectly (AUC 0.986 [95% CI 0.966–1.000]).

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Supplementary Material

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Glossary

AD = Alzheimer disease; **ADNC** = AD neuropathologic change; **AUC** = area under the curve; **BD-tau** = brain-derived tau; **FC** = fold change; **GFAP** = glial fibrillary acid protein; **iAD** = iatrogenic AD; **iCJD** = iatrogenic Creutzfeldt-Jakob disease; **IPD** = inherited prion disease; **IQR** = interquartile range; **NfL** = neurofilament light chain; **NHNN** = National Hospital for Neurology and Neurosurgery; **PRNP** = prion protein gene; **ROC** = receiver operating characteristic; **sAD** = sporadic AD; **sCJD** = sporadic CJD; **vCJD** = variant CJD.

Discussion

Plasma p-tau217 and p-tau181 are increased in both AD and prion diseases (regardless of burden of AD copathology). Diagnosing AD with a single abnormal p-tau plasma biomarker risks misdiagnosing prion diseases as AD. Plasma NfL/p-tau217 discriminates near-perfectly and could act as a flag to suspect prion diseases where this is a diagnostic possibility.

Classification of Evidence

This study provides Class II evidence that plasma NfL/p-tau217 discriminates patients with CJD from those with AD.

Introduction

The development of ultrasensitive plasma biomarker assays that measure proteins involved in Alzheimer disease (AD) pathobiology including phosphorylated tau (p-tau) and β -amyloid ($A\beta$) has created optimism that these tests could revolutionize AD diagnosis by enabling noninvasive, cost-efficient biomarker confirmation of underlying AD neuropathologic change (ADNC).^{1,2} Alzheimer's Association diagnostic criteria advocate separating clinical and biological diagnosis. They suggest a single abnormal "Core 1" biomarker of p-tau or $A\beta$ (including accurate plasma biomarkers such as p-tau217) would be sufficient to establish a diagnosis, even in asymptomatic patients.¹ However, little is known about how these tests perform in prion diseases, which can mimic AD clinically, and early studies show contrasting results.^{3,4}

Human prion diseases are fatal transmissible neurodegenerative diseases with a shared mechanism of formation of amyloid fibrillar assemblies of misfolded cellular prion protein, which propagate by elongation and fission.⁵ They cause approximately 1 in 5,000 deaths in countries where they are monitored well.⁶ Sporadic Creutzfeldt-Jakob disease (CJD), the most common human prion disease, is thought to arise from spontaneous formation of prions. Inherited prion diseases (IPDs) occur because of prion protein gene (*PRNP*) coding alterations. Acquired prion diseases arise from transmission of bovine spongiform encephalopathy prions (variant CJD) or human-human transmission through medical procedures (iatrogenic CJD).

Prion diseases including sporadic Creutzfeldt-Jakob disease (sCJD) are usually associated with rapid neurocognitive decline, but initial presentation as a pure cognitive syndrome is common,⁷ and in such cases, progression is typically slower.⁸ Diagnosis and referral to specialist services is often delayed until other symptoms and signs emerge by which time patients are severely impaired,⁹ despite advanced diagnostic

tests (MRI and real-time quaking-induced conversion) which otherwise permit early and accurate diagnosis.¹⁰ This presents a problem as rationally designed therapeutics against prion disease enter clinical testing,^{11,12} because meaningful treatment benefits are most likely in the early stages.

This study aimed to determine false positive rates for AD plasma biomarker tests in prion diseases and explore mitigations for blood-based biomarker diagnosis of neurodegenerative diseases. Plasma p-tau species are of particular concern because, although reported as highly accurate plasma biomarkers of AD, we predicted they would also be elevated in prion diseases, because hyperphosphorylated tau deposition is a neuropathologic feature of prion diseases^{13,14} and plasma and CSF total tau, p-tau181, and p-tau217 have, to varying extents, been reported to be elevated in prion diseases.^{3,4,13,15-19} In this study, we evaluated plasma p-tau217, p-tau181, $A\beta_{42/40}$ ratio, neurofilament light chain (NfL), brain-derived tau (BD-tau), and glial fibrillary acid protein (GFAP) from healthy controls, patients with AD, and patients with prion disease in a prospective natural history cohort including all types of prion disease, using postmortem cases to directly assess AD copathology. Our research questions were as follows: how do AD plasma biomarkers perform in discriminating AD and prion diseases? and what are the relationships between these biomarkers and prion disease progression measured directly using validated clinimetrics?^{9,20}

Methods

Participants and Plasma Samples

Individuals were eligible if they had enrolled in the National Prion Monitoring Cohort study⁹ or the PRION-1 trial,²¹ collectively known as "the Cohort," and had an EDTA plasma sample available. These included patients diagnosed by contemporary diagnostic criteria²² with sCJD, variant CJD (vCJD), iatrogenic Creutzfeldt-Jakob disease (iCJD) (secondary to cadaveric human growth hormone exposure), slow IPD (6-OPRI,